

Solutions to Dudley's *Elementary Number Theory*

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This article will contain solutions to many problems from Underwood Dudley's *Elementary Number Theory*. I picked problems that I thought had interesting solutions or proofs, demonstrated the content discussed in the chapter well, or just piqued my curiosity. The problems are from chapters 1 - 22 and appendix A on induction. The solutions and proofs have not been verified by anyone but myself.

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Chapter 1

Exercise 2:

If d divides b , then $b = ak$, where k is an integer. If b divides c , then $c = br$. Plugging in ak for b , $c = akr$. Obviously, a divides akr .

Exercise 3:

If d divides a , then a/d equals an integer k . It then follows that $a = dk$. So d divides ca can be rewritten, removing the a for dk , to $\frac{ca}{d} = \frac{cdk}{d}$. From the problem, ca is divisible by d , and cdk is also divisible by d .

Theorem 1:

Since d is the greatest common divisor of (a,b) , then no other greater number divides both a and b . If it did, d would no longer be the greatest common divisor. Since no other number can divide both, their greatest common factor is 1.

Problem 5:

$$N - abc = 1$$

The greatest common divisor must be 1 for (N,abc) and thus (N,a) , (N,b) , and (N,c) . If 1 is not the greatest common divisor, you would have an integer times an integer larger than one, that equals 1, which is impossible. Formal proof of that below.

If $\frac{N}{d} = k$ and $\frac{a}{d} = r$, then $N - abc = 1 = dk - drbc \leftrightarrow d(k-rbc) = 1$. d must then be equal to 1, or else the equation is impossible with two integers.

Problem 7:

$$\frac{b}{a} = k \leftrightarrow b = ak$$

$$\frac{a}{b} = r \leftrightarrow \frac{a}{ak} = r \leftrightarrow a = akr - \text{So } k \text{ and } r \text{ must equal } 1 \text{ or } -1 \text{ in order for } a = a \text{ to be true. So}$$

$$b = ak \leftrightarrow b = a(\pm 1)$$

Problem 8:

a is the largest number divisible by a , and it also divides b . Therefore, it must be the greatest common factor. If any larger number divides b , then it does not divide b , as it is larger than a .

Problem 9:

(a,b) is the largest number that can divide (a,b) and it also divides b by definition.

Problem 10:

We know that $n + (n + 1)$ must be divisible by the greatest common divisor. $n + (n + 1) = 2n + 1$, so the greatest common divisor must divide one, and the only integer that satisfies that is 1.

Problem 11a:

$k + (n + k) = dx$. where d is the greatest common divisor and x is a rate.

$k + (n + k) = dx \leftrightarrow 2k - dx = n$. Therefore, the greatest common divisor of k divides dx and n . Since $(k, n) = 1$, then the greatest common divisor must be one. If any other number divides k and n , then it also divides, or is, d , and the greatest common factor is not one.

Problem 12:

$b = ak$, $d = cr$, so $\frac{ac}{db} = \frac{ac}{akcr}$. Thus, ac divides bd .

Problem 13:

Since d divides a and b , then $a = dr$, $b = dk$. Thus, $\frac{ab}{d^2} \leftrightarrow \frac{drdk}{d^2} \leftrightarrow \frac{d^2kr}{d^2}$. So d^2 divides both.

Problem 14:

$\frac{ab}{c} = k \leftrightarrow b = \frac{ck}{a}$, and $\frac{a}{d} = x \leftrightarrow a = xd$. Then

$\frac{db}{c} \leftrightarrow \frac{dck}{ca}$, and c divides db .

Problem 15a:

$x^2 + ax + b = 0 \leftrightarrow x(x + a) = -b \leftrightarrow x + a = \frac{-b}{x}$.

x is the integer solution, so if $x + a$ is an integer, then $\frac{-b}{x}$ also must be an integer. So x , the solution, must divide b .

Chapter 2

Problem 5:

$$\sqrt{n} = \sqrt{p_1^{k_1} p_2^{k_2} p_3^{k_3} p_4^{k_4} \dots p_k^{k_x}} = p_1^{k_1/2} p_2^{k_2/2} p_3^{k_3/2} p_4^{k_4/2} \dots p_k^{k_x/2}.$$

Thus, every factor of p must be divisible by 2 to be a square. A number that is divisible by two is even, so n has all even exponents on its powers.

Problem 6:

Since the number x has even exponents, it can be split into two equal parts.

$a^{k_1} b^{k_2} \dots e^{k_x} \leftrightarrow (a^{k_1/2} b^{k_2/2} \dots e^{k_x/2})(a^{k_1/2} b^{k_2/2} \dots e^{k_x/2})$. If we say that a number $y = (a^{k_1/2} b^{k_2/2} \dots e^{k_x/2})$, then $y^2 = a^{k_1} b^{k_2} \dots e^{k_x} = x$. So x is a square with a root of y .

Problem 8:

No, consider if $d = ak$. Example $\frac{4*2}{8}$, $8(d)$ does divide $4(a)$ times $2(b)$, but it does not divide each of them separately.

Problem 9:

No, $n = p_1 p_2 \dots p_k$

So if a prime p_e divides n ,

$\frac{n}{p_e} = \frac{p_1 p_2 \dots p_k}{p_e}$, then that same prime divides $n + 1$.

$\frac{n+1}{p_k} = \frac{p_1 p_2 \dots p_l + 1}{p_k}$. So p_k has to divide 1, which implies that p_k does not exist.

Problem 10:

n and $n + 1$ do not have the same prime power compositions. A square, let's call ours x^2 , must have all even exponents. One of n or $n + 1$ must be even, as they are consecutive integers. So if $n = p_1 p_2 p_3 \dots p_k$ and $n + 1 = q_1 q_2 q_3 \dots q_e$. One of n or $n + 1$ has an odd exponent. This is

because the shortest even to even jump is from 1^2 to 2^2 , which is still 3; 2 more than the 1 integer apart that the consecutive integers have. An odd number exponent times an even number exponent always equals an odd number exponent, so it is proved.

Problem 13:

Consider the following situation,

$$n^{1/4} < p < q, pq = n^{3/4}$$

If $\frac{n}{pq}$ is a prime then $\frac{n}{pq} = p_2$, $n = p_2 pq \leftrightarrow p_2 n^{3/4}$ so p_2 must equal $n^{1/4}$. This implies that

$n = p_2 p_2 p_2 p_2$. Every number has only one prime factorization, but this result says that we have two, the original pqp_2 and $p_2 p_2 p_2 p_2$. This is a contradiction, and proves the problem false.

Problem 14:

$2^{n-1} = 2(2)(2)(2)...$ until $n-1$ equals zero. Since an even number is divisible by two, the only value this is non composite for is $n = 1$ or 2 .

Chapter 3

Problem 5:

$$x + y + z = 41; x + 2y + 3z = 31.$$

From this, $ax + by + cz = d$. Then we take a second solution, called (r, k, s) .

$$a(x - r) + b(y - k) + c(z - s) = d - d, a \text{ divides } (y - k), \text{ so}$$

$$am = y - k \leftrightarrow k = y - am. a \text{ also divides } (z - s), \text{ so } at = z - s \leftrightarrow s = z - at. \text{ So}$$

$$a(x - r) + b(y - k) + c(z - s) = 0 \leftrightarrow a(x - r) + bam + cat = 0 = x - r + bm + ct \leftrightarrow r = x + bm + ct.$$

Solution found by inspection is $(47, -2, -4)$ so

$$k = y - am \leftrightarrow k = -2 - m$$

$$s = z - at \leftrightarrow -4 - t$$

$$r = x + bm + ct$$

Then we must set this equal to the first equation, so

$$-2 - m - 4 - t + 47 + 2m + 3t = 41 \leftrightarrow m = -2t$$

Infinite solutions can now be generated using these equations.

Problem 6:

$$p + d + q = 100, p + 10d + 25q = 499$$

A solution found by inspection is $(99, 30, 4)$.

$$ap + bd + cq = d = ax + by + cz \leftrightarrow d - d = a(p - x) + b(d - y) + c(q - z)$$

a divides $(d - y)$ and $(q - z)$ because it is relatively prime with b and c . So

$am = d - y \leftrightarrow y = d - am$ and $at = q - z \leftrightarrow z = q - at$. Plugging this in to the prior equation we get $a(p - x) + b(d - y) + c(q - z) \leftrightarrow x = p + bm + ct$. Using the solution,

$$y = d - am \leftrightarrow y = 30 - m$$

$$z = q - at \leftrightarrow z = 4 - t$$

$$x = p + bm + ct \leftrightarrow 99 + 10m + 25t$$

This must also equal 100 for the answer to be valid for both equations. So

$30 - m + 4 - t + 99 + 10m + 25t = 100 \leftrightarrow 3m + 8t = -11$. Infinite answers can be generated using this relationship. For example $m = -1, t = -1$ yields $(64, 31, 5)$, which is an answer

Problem 9:

If x is the first merchant, y is the second merchant, z is the third merchant, and p is the purse, then this equation can be written as

$$x + p = 2(y + z)$$

$$y + p = 3(x + z)$$

$$z + p = 5(x + y)$$

$$z + p = 5x + 5y \leftrightarrow 2z + 2p = 10(x + y)$$

From the first equation,

$$2z = x + p - 2y, \text{ so}$$

$$x + p - 2y + 2p = 10x + 10y \leftrightarrow 3p = 9x + 12y \leftrightarrow p = 3x + 4y$$

From the first equation again,

$$2z = x + p - 2y \leftrightarrow 2z = x + (3x + 4y) - 2y \leftrightarrow z = 2x + y$$

Using the second equation,

$$y + p = 3(x + (2x + y)) \leftrightarrow y + p = 9x + 3y \leftrightarrow p = 9x + 2y \leftrightarrow 3x + 4y = 9x + 2y \leftrightarrow y = 3x$$

From here, it is easy to plug back in and find out that:

$$x = d$$

$$y = 3d$$

$$z = 5d$$

$$p = 15d$$

Where d can be any integer. A better way would be to use a matrix method and reduce it, but this way works fine as well.

Chapter 4

Exercise 5:

$$a = r(\text{mod } p) \leftrightarrow a = pk + r \leftrightarrow \frac{a}{p} = k + \frac{r}{p}$$

So the only way that p divides a is if p divides r , which means $a \equiv 0(\text{mod } p)$

Exercises 6, 7, 8, and 9:

$$\text{b. } a = q_1 m + r \text{ and } b = q_2 m + r$$

$$b - a = km = q_2 m + r - q_1 m - r \leftrightarrow m(q_2 - q_1) = km$$

This last statement is the definition of a modulus.

$$\text{c. } a - b = km \leftrightarrow a = b + km$$

$$b - c = k_1 m \leftrightarrow c = b - k_1 m$$

$$a \equiv c \leftrightarrow b + km \equiv b - k_1 m \pmod{m} \leftrightarrow b + km - b + k_1 m \leftrightarrow m(k + k_1)$$

This is the definition of a modulus.

$$\text{d. } a = b + km \text{ and } c = d + k_1 m$$

$$\text{So } a + c = b + d + m(k + k_1) \equiv b + d \pmod{m}$$

$$b + d + m(k + k_1) - b - d = rm \leftrightarrow m(k + k_1) = rm$$

Again, this is the definition of a modulus.

Problem 3:

$$\text{Since } a \equiv b \pmod{m}, a = q_1 m + r \text{ and } b = q_2 m + r$$

$$a^2 = q_1^2 m^2 + 2q_1 mr + r^2$$

$$b^2 = q_2^2 m^2 + 2q_2 mr + r^2$$

$$a^2 - b^2 = m(q_1^2 m^2 + 2q_1 r - q_2^2 m - 2q_2 r)$$

$$\text{Since } a^2 - b^2 \text{ equals } m \text{ times a rate, then } a^2 \equiv b^2 \pmod{m}.$$

Problem 8:

A prime (mod 4) cannot be congruent to 0 because then it would be divisible by 4. It also cannot be congruent to 2, because then it would be divisible by two. $a = q_1 4 + 2 \leftrightarrow a = 2(q_1 2 + 1)$

Therefore, the prime must be congruent to one or three.

Problem 9:

Like above, the prime cannot have a least residue of 0 or 6 or else it is divisible by 6. For the other situations:

If congruent to 2, $a = q_1 6 + 2$; divisible by 2.

If congruent to 3, $a = q_2 6 + 3$; divisible by 3.

If congruent to 4, $a = q_3 6 + 4$; divisible by 4.

Thus the prime can only be congruent to 1 or 5.

Problem 7:

$$k = q_1 4 + 1, \text{ so } 6k + 5 \leftrightarrow 6(q_1 4 + 1) + 5 \leftrightarrow 24q_1 + 11$$

$$24q_1 + 11 \leftrightarrow 24q_1 + 8 + 3 \leftrightarrow 4(6q_1 + 2) + 3 \leftrightarrow 6q_1 + 2 \equiv 3 \pmod{4}$$

So the number is congruent to 3 modulo 4.

Problem 10:

A prime must be so that $(p, 30) = 1$. Thus any prime below 30 could be congruent to a prime.

Problem 13:

Every square has as its last digit 0, 1, 4, 5, 6, or 9. Since the last digit is the remainder of these numbers modulo 10:

$$0 * 0 \equiv 0$$

$$1 * 1 \equiv 1$$

$$4 * 4 = 6$$

$$5 * 5 = 5$$

$$6 * 6 = 6$$

$$9 * 9 = 1$$

So a fourth power can have 0, 1, 5, or 6 as its last digit.

Problem 14:

$$\frac{a^3 - (a+1)^3}{3} = k, \text{ where } k \text{ is an integer.}$$

Expanding and multiplying by 3,

$$a^3 - a^3 - 3a^2 - 3a + 1 = 3k \leftrightarrow -3a^2 - 3a + 1 \leftrightarrow -a^2 - a + \frac{1}{3} = k$$

This is a contradiction, as an integer cannot equal two integers plus a fraction. Also, only every third number is divisible by three, so it isn't possible for two consecutive integers to be divisible by three. You can use this same logic to prove problem 15 as well.

Problem 18a:

For this proof, first prove that every other 10^x is congruent modulo 11.

If 10^x and 10^{x+2} are congruent to each other modulo 11, then the difference must be divisible by 11. So

$$10^{x+2} - 10^x = 11k \leftrightarrow 10^2 * 10^x - 10^x = 11k \leftrightarrow 10^x(100 - 1) = 11k \leftrightarrow 10^x * 9 * 11 = 11k.$$

So the difference is divisible by 11, thus they are congruent modulo 11. Armed with this knowledge we can prove this problem and problem 18b.

An palindrome $abba$ where a and b are the actual digits, can be represented by:

$$\frac{10^3 a + 10^2 b + 10 b + a}{11} = k$$

Since we know that every other power of ten has the same residue (mod 11), we can easily simplify this problem. a leaves a remainder of a when divided by 11, so 10^2b leaves a remainder of b modulo 11. Similarly, $10^3a = 90a(11) + 10a$, so 10^3a leaves a remainder of $10a$. Now we know that $10b$ also leaves a remainder of $10b$ when divided by 11, so the equation can be simplified down to:

$$\frac{10a+b+10b+a}{11} \leftrightarrow \frac{11(a+b)}{11} = 11k$$

Since it is divisible by 11, it is proved.

Problem 18b:

A palindrome $abccba$ where a, b, c are the digits, can be shown as

$$10^5a + 10^4b + 10^3c + 10^2c + 10b + a = 11k.$$

From the proof above, it's known that $a, 10^2c, 10^4b$ all leave a remainder of a, b, c respectively when divided by 11. It's also known that $10b, 10^3c, 10^5a$ leave a remainder of $10(a, b, c)$ when divided by 11. So

$$\frac{10^5a+10^4b+10^3c+10^2c+10b+a}{11} \leftrightarrow \frac{10a+b+10c+c+10b+a}{11} = \frac{11(a+b+c)}{11} = 11k. \text{ Both are divisible by 11,}$$

so the problem is proven.

Problem 19:

$$n \equiv 4 \pmod{9} \leftrightarrow n = 9q + r \leftrightarrow n^3 = (9q + r)^3$$

So the remainder is simply $r^3 \pmod{9}$.

$$0^3 = 0, 1^3 = 1, 2^3 = 8, 3^3 = 0, 4^3 = 1, 5^3 = 8, 6^3 = 0, 7^3 = 1, 8^3 = 8, 9^3 = 0.$$

You cannot make a number congruent to 4 modulo 9 with these numbers.

Problem 20:

$$x \equiv 1 \pmod{m^k} \leftrightarrow x = am^k + 1 \leftrightarrow x^m = (am^k + 1)^m.$$

All of the terms in $(am^k + 1)^m$ are immediately seen as divisible by am^{k+1} , besides the last two terms. The second to last term will be in the form bam^k but b actually equals m , as however many b 's there are is simply the value the equation is raised to, m . So $bam^k \leftrightarrow mam^k \leftrightarrow am^{k+1}$.

So the entire equation besides the last term is divisible by am^{k+1} . The last integer is in the form $1^m = 1$, so $x^m \equiv 1 \pmod{m^{k+1}}$ is a true statement

Chapter 5

Problem 3b:

$$x \equiv 3 \pmod{5} \leftrightarrow x = 5k_1 + 3$$

$$x \equiv 5 \pmod{7} \leftrightarrow 5k_1 + 3 \equiv 5 \pmod{7}$$

$$\text{A solution is } 5k_1 + 3 \equiv 33 \pmod{7} \leftrightarrow 5k_1 - 30 = 7s \leftrightarrow 5(k_1 - 6) = 7s$$

$$7 \text{ must divide } (k_1 - 6) \text{ so } 7k_2 = k_1 - 6 \leftrightarrow k_1 = 6 + 7k_2$$

Plugging back in for x:

$$x = 35k_2 + 33$$

$$35k_2 + 33 \equiv 7 \pmod{11} \leftrightarrow 11(3k_2 + 3) + 2k_2 \equiv 7 \pmod{11} \leftrightarrow 2k_2 \equiv 7 \pmod{11}$$

$$\text{A solution to this is } 2k_2 \equiv 18 \pmod{11} \leftrightarrow k_2 = 11k_3 + 9$$

Plugging back in for x for the final time:

$$x = 385k_3 + 348 \text{ or } x \equiv 348 \pmod{385}$$

Problem 3c:

$$2x = 5k_1 + 1 \leftrightarrow 6x = 15k_1 + 3$$

$$15k_1 + 3 \equiv 4 \pmod{7} \leftrightarrow 15k_1 - 1 = 7k_2$$

$$(7 * 2)k_1 + k_1 - 1 = 7k_2 \leftrightarrow k_1 \equiv 1 \pmod{7} \leftrightarrow k_1 = 7k_2 + 1$$

$$\text{So } 2x = 35k_2 + 6 \text{ and } 4x = 70k_2 + 12$$

$$70k_2 + 12 \equiv 2 \pmod{11}$$

$$11(6k_2 + 1) + 4k_2 - 2 = 11k_3 \leftrightarrow 4k_2 \equiv 2 \pmod{11k_3}$$

$$\text{So } 4k_2 \equiv 24 \pmod{11} \leftrightarrow k_2 \equiv 6 \pmod{11} \leftrightarrow k_2 = 11k_3 + 6 \text{ and}$$

$$2x = 385k_3 + 216 \leftrightarrow 2x \equiv 216 \pmod{385} \leftrightarrow x \equiv 108 \pmod{385}$$

We can times the congruences freely because:

$$a \equiv b \pmod{m} \leftrightarrow ax \equiv bx \pmod{m}$$

$$ax - bx = mk \leftrightarrow x(a - b) = mk$$

So mk divides a-b or else the first congruence is false. The book has a more concise solution, $x \equiv 3 \pmod{385}$, but 108 works as well.

Problem 4a:

$$x = 2k_1 + 1 \leftrightarrow 2k_1 + 1 \equiv 2 \pmod{3} \leftrightarrow 2k_1 - 1 = 3m$$

$$2k_1 \equiv 1 \pmod{3} \text{ so } 2k_1 \equiv 4 \pmod{3} \leftrightarrow k_1 \equiv 2 \pmod{3} \leftrightarrow k_1 = 3k_2 + 2 \leftrightarrow x = 6k_2 + 5$$

$$\text{Thus } x \equiv 5 \pmod{6}$$

Problem 9a:

$x + 2y \equiv 3 \pmod{7} \leftrightarrow x = 7k_1 + 3 - 2y$ so:

$3x + y \leftrightarrow 21k_1 + 9 - 5y = 7k_2 + 2 \leftrightarrow -5y + 7 = 7(k_2 - 3k_1)$, a solution is

$y \equiv 0 \pmod{7} \leftrightarrow y = k_2 - 3k_1$; so x then equals

$7k_1 + 3 - 2(k_2 - 3k_1) = x = 3 + k_1 - 2k_2$. The basic solution at $k_1 = 0, k_2 = 0$ is $x = 3$ and $y = 0$.

Problem 11

This problem can be written as

$$x = 4k_1 + 1$$

$$x \equiv 1 \pmod{4}$$

$$x \equiv 2 \pmod{5}$$

$$x \equiv 3 \pmod{7} \text{ so}$$

$$4k_1 + 1 = 5k_2 + 2 \leftrightarrow 4k_1 - 1 = 5k_2 \leftrightarrow 4k_1 \equiv 1 \pmod{5} \leftrightarrow 4k_1 \equiv 16 \pmod{5}$$

$$k_1 \equiv 4 \pmod{5} \leftrightarrow k_1 = 5k_2 + 4$$

$$x = 20k_2 + 17$$

$$20k_2 + 17 - 3 = 7k_3 \leftrightarrow 7(2k_2 + 2) + 6k_2 = 7k_3$$

$k_2 \equiv 0 \pmod{7} \leftrightarrow k_2 = 7k_3$. Plugging in, $x = 140k_3 + 17$. Possible solutions include 17, 157, 297...

Problem 14:

This problem can be written as

$$n \equiv 2 \pmod{3}$$

$$n \equiv 2 \pmod{4}$$

$$n \equiv 2 \pmod{5}$$

$$n \equiv 2 \pmod{6}$$

$$n = 2k_1$$

$$\text{So } 2k_1 \equiv 2 \pmod{3} \leftrightarrow k_1 \equiv 1 \pmod{3} \leftrightarrow k_1 = 3k_2 + 1$$

$$n = 6k_2 + 2$$

$$6k_2 + 2 - 2 = 4k_3 \leftrightarrow k_2 = 4k_3 \text{ so } n = 24k_3 + 2$$

$$24k_3 + 2 - 2 = 5k_4 \leftrightarrow k_3 = 5k_4. \text{ Plugging in for } n:$$

$$n = 120k_4 + 2$$

This already satisfies the last congruence, so I will omit the extra work. The smallest possible answer is 2.

Problem 15:

The integer must be written in the form $2^{k_1} 3^{k_2} 5^{k_3}$ to solve with congruences. Half of the integer is a square so $k_1 - 1 = 2a \leftrightarrow k_1 \equiv 1 \pmod{2}$. k_1 also has no remainder when divided by three because it must be a cube. So $k_1 \equiv 0 \pmod{3}$ and it also must be divisible by five;

$k_1 \equiv 0 \pmod{5}$. A method is to continue adding 2 to $k_1 \equiv 1 \pmod{2}$ until the congruence equals an integer that is divisible by 3 and 5. The first one is

$k_1 \equiv 15 \pmod{2} \leftrightarrow k_1 = 2a + 15$. The same method can be used to find $k_2 = 3b + 10$ and

$k_3 = 5c + 6$. The smallest number is $2^{15} * 3^{10} * 5^6$. Another such integer is $2^{45} * 3^{40} * 5^{36}$.

Problem 16a:

$$n = 9k_1; n + 1 = 16k_2; n + 2 = 25k_3$$

$9k_1 \equiv -1 \pmod{16}$ solving for k_1 ,

$$k_1 = 16k_2 + 7 \leftrightarrow n = 144k_2 + 63 \leftrightarrow 144k_2 + 63 \equiv -2 \pmod{25}. \text{ Solving for } k_2,$$

$$k_2 = 25k_3 + 15 \leftrightarrow n = 3600k_3 + 2223. \text{ So the smallest number is } 2223.$$

Problem 17:

I am going to prove that if $x - rm - r + sm = m(m + 1)k$, then x is congruent to r modulo m and x is congruent to s modulo $m + 1$. This is the opposite of what the problem asked us to prove, but it is essentially the same thing.

$$x - rm - r + sm = m(m + 1)k \leftrightarrow x - r + m(s - r) = m(m + 1)k.$$

So m must divide $x - r$ and $x \equiv r \pmod{m}$

$$x - r = qm \leftrightarrow qm + m(s - r) = m(m + 1)k \leftrightarrow q + s - r = (m + 1)k$$

$$-r = qm - x \text{ so } q + s - x + qm = (m + 1)k \leftrightarrow q(1 + m) + s - x = (m + 1)k.$$

$m + 1$ divides $s - x$, or $x \equiv s \pmod{m + 1}$.

Another way to solve this is to reduce the congruences and least residues until you get

$$\frac{x}{m(m+1)} - \frac{x}{m(m+1)} = k, \text{ which is merely asking if } x \text{ is congruent to itself.}$$

Problem 18:

$$3x = 20a + a = 21a$$

$$5y = 20(a + 1) + (a + 1) = 21(a + 1)$$

$$7z = 20(a + 2) + (a + 2) = 21(a + 2)$$

Factoring out the 21, $x = a$, $y = a + 1$, and $z = a + 2$. But they must be divisible by 3, 5, and 7 to satisfy the congruences.

$$3k_1 \equiv -1 \pmod{5} \leftrightarrow 3k_1 \equiv 4 \pmod{5} \leftrightarrow 3k_1 \equiv 9 \pmod{5} \leftrightarrow k_1 \equiv 3 \pmod{5} \leftrightarrow k_1 = 5k_2 + 3.$$

$$a = 15k_2 + 9$$

$$15k_2 + 11 = 7k_3 \leftrightarrow 7(2k_2 + 1) + k_2 + 4 = 7k_3 \leftrightarrow k_2 \equiv -4 \pmod{7}$$

$$k_2 \equiv 3 \pmod{7} \leftrightarrow k_2 = 7k_3 + 3$$

$$15(7k_3 + 3) + 9 = 105k_3 + 54.$$

So any integer of the form $105k_3 + 54$ will satisfy for a . The smallest answer is $x = 18$, $y = 11$, and $z = 8$.

Appendix A

I figure that right now is a good interjection point to show some answers for Appendix A on induction. Induction will only become more relevant as the book goes on, so it is best to make sure one has a grasp of the subject.

Problem 1:

$$0^2 = \frac{0}{6} = 0.$$

The equation must get to the form $1^2 + 2^2 \dots + n^2 + (n + 1)^2 = \frac{(n+1)(2n+3)(n+2)}{6}$

$$1^2 + 2^2 \dots n^2 + (n + 1)^2 = \frac{(n)(n+1)(2n+1)}{6} + (n + 1)^2 \leftrightarrow \frac{(n)(n+1)(2n+1) + 6(n+1)^2}{6} \leftrightarrow \frac{(n+1)(n(2n+1) + 6(n+1))}{6}$$

$$\frac{(n+1)(2n^2 + 7n + 6)}{6} \leftrightarrow \frac{(n+1)(2n+3)(n+2)}{6}$$

Problem 2:

I guessed the formula $1^3 + 2^3 \dots n^3 = (\frac{n(n+1)}{2})^2$. So we are trying to prove

$$1^3 + 2^3 \dots a^3 + (n+1)^3 = (\frac{(n+1)(n+2)}{2})^2$$

$$1^3 + 2^3 \dots a^3 + (n+1)^3 = (\frac{n(n+1)}{2})^2 + \frac{4(n+1)^3}{4} \leftrightarrow \frac{(n+1)^2(n^2+4n+4)}{4} \leftrightarrow \frac{(n+1)^2(n+2)^2}{2^2} \leftrightarrow (\frac{(n+1)(n+2)}{2})^2$$

Problem 4:

$$\text{Trying to prove } \frac{1}{1*2} \dots + \frac{1}{(n-1)(n)} + \frac{1}{(n+1)(n)} = 1 - \frac{1}{n+1}$$

$$\text{The sequence plus } \frac{1}{(n+1)(n)} = 1 - \frac{1}{n} + \frac{1}{(n+1)(n)}$$

$$1 + \frac{1}{(n+1)(n)} - \frac{1(n+1)}{n(n+1)} = 1 + \frac{-n}{n(n+1)} = 1 - \frac{1}{n+1}$$

Problem 6:

$$\text{If } a_n = 2^n - 1, \text{ then } a_{n+1} = 2^{n+1} - 1.$$

Since a_{n+1} is set equal to a function containing a_n , you can just substitute a_n in.

$$a_{n+1} = 2(2^n - 1) + 1 \leftrightarrow a_{n+1} = 2^{n+1} - 2 + 1 \leftrightarrow a_{n+1} = 2^{n+1} - 1. \text{ So it is proven.}$$

Problem 7:

$$\text{We need } a_{n+1} = \frac{2^{n+2} + (-1)^{n+1}}{3}$$

$$a_n = \frac{2^{n+1} + (-1)^n}{3}$$

$$\text{Using that } a_{n+1} = a_n + 2a_{n-1}, a_n = \frac{2^{n+1} + (-1)^n}{3} \leftrightarrow a_{n+1} = \frac{2^{n+1} + (-1)^n}{3} + 2a_{n-1}$$

$$\text{If the induction statement is true, then } a_{n-1} = \frac{2^n + (-1)^{n-1}}{3}, \text{ so:}$$

$$a_{n+1} = \frac{2^{n+1} + (-1)^n}{3} + \frac{2*2^n + 2(-1)^{n-1}}{3} = \frac{2^{n+1} + 2^{n+1} + (-1)^n + 2(-1)^{n-1}}{3} = \frac{2(2^{n+1}) + (-1)^n(1-2)}{3} = \frac{2^{n+2} + (-1)^{n+1}}{3}$$

And it is proven.

Problem 8:

Since the greatest common factor is one, there are numbers x and y such that

$$x(a_n) + y(a_n + 1) = 1. \text{ To prove it, the equation must be in the form}$$

$k_1(a_n + 1) + k_2(a_n + 2) = 1$. The a_n needs to go, and since $a_{n+2} = 3a_{n+1} + a_n$ we can substitute for a_n in terms of a_{n+2} and a_{n+1} . So:

$$x(a_n) + y(a_{n+1}) = 1 = x(a_{n+2} - 3a_{n+1}) + y(a_{n+1}) = 1$$

Distributing the x and factoring out the a_{n+1} to get in the form $a_{n+1}(y - 3x) + a_{n+2}(x) = 1$, which is the same form as $k_1(a_{n+1}) + k_2(a_{n+2}) = 1$. They are both in the same form, just with differing variables.

Problem 9:

The theorem I have guessed is $(5 + 4n + 2(\frac{(n+1)(n)}{2}))^2$, which reduces to

$$(5 + 5n + n^2)^2 - 1 = \frac{(n+4)!}{n!}$$

$$\frac{n+5}{n+1} * (5 + 5n + n^2)^2 - 1 = \frac{(n+4)*(n+5)!}{n!*(n+1)} \leftrightarrow \frac{n+5}{n+1} * (5 + 5n + n^2)^2 - \frac{n+5}{n+1} = \frac{(n+5)!}{(n+1)!}$$

If you expand this fully, you get $n^4 + 14n^3 + 71n^2 + 154n + 120$, which is what the desired equation also equals when expanded out. I could not find a more elegant way to simplify and factor the equation into the desired form.

Problem 11:

Expanding both $n(n + 1)(n + 2)$ and $(n + 1)(n + 2)(n + 3)$, we find that the latter is $3(n + 2)(n + 1)$ bigger than the former. This is a multiple of six, since one of the n s must be odd, which then creates an even number, and $2 * 3 = 6$. So:

$$n(n + 1)(n + 2) + 3(n + 2)(n + 1) = 6k_1 + 3(n + 2)(n + 1) \leftrightarrow$$

$$(n + 1)(n + 2)(n + 3) = 6(k_1 + k_2). \text{ The form is correct, and it is divisible by 6.}$$

Problem 12:

$$\frac{17}{24} (x - 1)(x - 2)(x - 3)(x - 4)$$

Problem 14:

Instead of proving this one by induction, I will prove it in what I consider a much easier way, using factoring and cases.

$$n^4 - n \leftrightarrow n(n^4 - 1) \leftrightarrow n(n^2 + 1)(n^2 - 1) \leftrightarrow n(n - 1)(n + 1)(n^2 + 1).$$

The only two numbers not immediately seen as divisible by 5 are 3 and 2. 3 squared plus one leaves a remainder of 0 modulo 5, and 2 squared also leaves a remainder of 0 modulo 5. Those are the only cases left, so they are all divisible by 5.

Chapter 6

Problem 3:

$7^{355} \pmod{10} - 7^1 \equiv 7; 7^2 \equiv 9; 7^4 \equiv 1$. So all other 7^x will also equal one.
 $7^{256} * 7^{64} * 7^{32} * 7^2 * 7^1 \equiv 1 * 9 * 7 \equiv 63 \equiv 3$. The last digit is three.

Problem 4:

$7^{355} \pmod{10} - 7^1 \equiv 7; 7^2 \equiv 49; 7^4 \equiv 1$. So
 $7^{256} * 7^{64} * 7^{32} * 7^2 * 7 \equiv 1 * 49 * 7 \equiv 343 \equiv 43 \pmod{10}$. So the two digits are 43.

Problem 9:

$$p - 1 = r$$

$$p * 2 = r - 1 \dots p - r = 1.$$

$$\text{So } (p - 1)(p - 2) \dots (p - r) \equiv r! \leftrightarrow (-1)^r r! \pmod{p} \leftrightarrow 1 \equiv (-1)^r \pmod{p}$$

Every prime besides two is even, so the above statement is true.

Problem 10:

Every non prime integer less than n divides $(n - 1)!$ Since n is not prime, it has factors other than itself and one that are smaller than it. $(n - 1)!$ will of course contain all of these factors, so it is divisible by n .

Problem 11:

$$-1 \equiv (p - 1)! \pmod{p} \leftrightarrow -1 \equiv (p - 1)(p - 2)(p - 3)! \leftrightarrow$$

$$-1 \equiv (p^2 - 3p + 2)(p - 3)! \leftrightarrow (p(p - 3) + 2)(p - 3)! \equiv -1 \pmod{p} \leftrightarrow$$

$$p(p-3) \equiv 0 \pmod{p} \text{ so } p(p-3) + 2 \leftrightarrow 0 + 2 \leftrightarrow 2(p-3)! \equiv -1 \pmod{p} \leftrightarrow 2(p-3)! + 1 \equiv 0 \pmod{p}$$

Problem 12:

$$n + 1 = p, \text{ so } n = p - 1$$

$$\text{A number } a^n \equiv a^{p-1}, \text{ and } b^n \equiv b^{p-1}$$

$$a^{p-1} \equiv 1 \pmod{p} \text{ and } b^{p-1} \equiv 1 \pmod{p} \text{ so } a^{p-1} \equiv b^{p-1} \pmod{p} \leftrightarrow a^{p-1} - b^{p-1} = pk$$

$a^n - b^n$ is divisible by a prime.

Problem 14:

The theorem I guessed is $(p-n)!(n-1)! \equiv (-1)^n$. To prove by induction we need to get to the form $(p-n-1)!(n-1)! \equiv (-1)^{n+1}$.

$$\begin{aligned} (-1)(p-n)(p-n-1)! &\equiv (-1)^n(-1) \leftrightarrow (n-p)(n-1)!(p-n-1)! \equiv (-1)^{n+1} \leftrightarrow \\ (n(n-1)! - p(n-1)!)(p-n-1)! &\equiv (-1)^{n+1} \leftrightarrow (n! - p(n-1)!)(p-n-1)! \equiv (-1)^{n+1} \\ p(n-1)! &\text{ is divisible by } p \text{ and leaves a remainder of } 0 \text{ modulo } 11. \end{aligned}$$

$$(n! - 0)(p-n-1)! \equiv (-1)^{n+1} \leftrightarrow (n!)(p-n-1)! \equiv (-1)^{n+1} \pmod{11}$$

Problem 15a:

Since $a^{p-1} \equiv 1 \pmod{p}$, $1^{p-1} + 2^{p-1} + 3^{p-1} \dots \equiv 1 + 1 + 1 \dots$ for a $p-1$ amount of times. So it equals $p-1$. $p-1 \equiv -1 \pmod{p}$ so it is proven.

Problem 15b:

$$a^p \equiv a \pmod{p} \text{ so } 1^p \equiv 1, 2^p \equiv 2 \dots, \text{ so}$$

$$1^p + 2^p + \dots + (p-1)^p \leftrightarrow 1 + 2 + 3 + \dots + (p-1) = \frac{(p-1)p}{2}, \text{ which is of course divisible by } p.$$

Problem 17a:

If the sequence is divisible by pq , then it also must be divisible by p and q separately. I am going to calculate the sequence's modulo p and q separately.

$$a^p \equiv a \pmod{p} \leftrightarrow a^p * a^q \equiv a^{q+1} \pmod{p}.$$

$$a^p \equiv a \pmod{p} \leftrightarrow a^{p+1} \equiv a^2 \pmod{p}.$$

$$\text{So } a^{p+q} - a^{p+1} - a^{q+1} + a^2 \leftrightarrow a^{q+1} - a^2 - a^{q+1} + a^2 \leftrightarrow 0 \equiv 0 \pmod{p}.$$

A similar method can be found to prove that the sequence is also congruent to 0 modulo p . This is sufficient to prove that the sequence is divisible by pq .

Problem 17b:

$a^p \equiv a \pmod{p}$ and $a^{pq} \equiv a^q \pmod{p}$, so

$$a^{pq} - a^p - a^q + a \leftrightarrow a^q - a - a^q + a \equiv 0 \pmod{p}$$

$a^q \equiv a \pmod{q}$ and $a^{pq} \equiv a^p \pmod{q}$ so

$$a^{pq} - a^p - a^q + a \leftrightarrow a^p - a^p - a + a \leftrightarrow 0 \equiv 0 \pmod{q}$$

Since we have proved that the sequence is divisible by p and q , it follows that it is divisible by pq .

Problem 18:

$a^{p-1} \equiv 1 \pmod{p} \leftrightarrow a^{p-1} * a^p \equiv a^p \leftrightarrow a^{2p-1} \equiv a^p \equiv a$. So

$2^{2p-1} \equiv 2 \pmod{p} \leftrightarrow 2 - 2 \equiv 0 \pmod{p}$, which is a true statement.

Problem 19:

$1 + n + n^2 \dots n^{p-2} = a$ where a is the sum of the sequence.

$$n(1 + n + n^2 \dots n^{p-2}) = an$$

$$an - a = (1 + n + n^2 \dots n^{p-2}) - (n + n^2 + n^3 \dots n^{p-1}) \leftrightarrow a(n - 1) = 1 - n^{p-1} \leftrightarrow a = \frac{1-n^{p-1}}{n-1}$$

.Since $n^{p-1} \equiv 1 \pmod{p}$, the sequence is divisible by p as long as n is not congruent to 1 or 0 modulo p . If the sequence is congruent to 1, then the denominator equals 0. If the sequence is congruent to 0, then the sequence is congruent to $1 + 0 + 0 \dots$, which equals 1.

Problem 20:

1111....111 can be written as $10^k + 10^{k-1} \dots 10^1 + 1$. This is in the same form as the above problem, so we know that a prime divides the form $10^{p-2} \dots 10^1 + 1$. Thus every prime divides a version of 1111...111.

Chapter 7

Exercise 7:

Every ϵ of the sequence has $n + 1$ divisors, and in terms of 2^x . So

$$2^n + 2^{n-1} \dots + 1 = a = \text{sum}$$

$$2^{n+1} + 2^n \dots + 2 = 2a$$

$$2a - a = a = (2^{n+1} + 2^n \dots + 2) - (2^n + 2^{n-1} \dots + 1) = 2^{n+1} - 1$$

It follows that the 2 actually equal to one of the 2^{n-x} that equals 2. So they cancel and leave the negative one.

Problem 5:

If n is a power of 2, then it follows that $\sigma(n) = 2^{n+1} - 1$. 2^{n+1} will always be even for $n > 0$, and one less than an even number is an odd number.

Problem 6:

$n = p_1^{e_1} p_2^{e_2} \dots p_x^{e_x}$. For $\frac{f(n)}{n}$ to be multiplicative, then

$$\frac{f(n)f(p_{x+1}^{e_{x+1}})}{n} = \frac{f(n \cdot p_{x+1}^{e_{x+1}})}{n}$$

$$\frac{f(n)f(p_{x+1}^{e_{x+1}})}{n} = \frac{f(p_1^{e_1})f(p_2^{e_2}) \dots f(p_{x+1}^{e_{x+1}})}{n}$$

Since $f(n)$ is multiplicative,

$$\frac{f(p_1^{e_1}) \dots f(p_{x+1}^{e_{x+1}})}{n} \leftrightarrow \frac{f(n \cdot p_{x+1}^{e_{x+1}})}{n}$$

Problem 7:

$$2^4 * 3^2 * 5 * 7$$

Problem 10:

For a prime composition of 2 integers ,

$$(x + 1)(y + 1) = 60$$

$$2^x * 3^y$$

For an undefined number

$$p_1^{a_1} p_2^{a_2} \dots p_x^{a_x} \text{ as long as } (a_1 + 1)(a_2 + 1) \dots (a_x + 1) = 60$$

An example generate from this is $5^4 * 7^3 * 13^2$

Problem 12:

$\sigma(n)$ is odd iff its prime factor exponents add up to an even number. If 2^x is included in the factors and the factor sum is even, then nothing is added to the sum. If it is odd, then 1 is added to the factors sum.

$$n = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k}$$

$$\sigma(n) = \sigma(p_1^{e_1}) \dots \sigma(p_x^{e_x})$$

It follows that $\sigma(p_x^{e_x}) = 1 + p_x + p_x^2 \dots p_x^{e_x}$. If there are an even number amount of p_x 's, then the sequence equals an odd number. This requires an even exponent. Thus $\sigma(p_x^{e_x})$ is odd iff e is an even number. Since $\sigma(2^x)$ is always odd, you do not add it to the exponent count. As an example, using this method it trivial to ask if $\sigma(2^{16} * 3^3 * 5^2 * 7^2)$ is odd or not. 2^{16} is odd, and does not impact the exponent count. 3^n is even, so $\sigma(n)$ is even. To generalize even further, $\sigma(n)$ will be odd iff it has no prime factors with no odd exponents.

Problem 13:

First, $d(p^e) = 1 + e$

$n = p_1^{e_1} p_2^{e_2} \dots p_x^{e_x}$ where e is even because n is a square. So:

$d(n) = d(p_1^{e_1}) \dots d(p_x^{e_x}) = (e_1 + 1) \dots (e_x + 1)$. Since, e is even, $e + 1$. An odd number to any power is still odd.

Problem 14:

Following the logic from the last problem, any n with $d(n) = 2a + 1$ has an even exponent on its prime factors. Of course any n with all even exponents is a square.

Problem 16:

If n is a prime other than 2 or 3, then $\sigma(n) = 1 + p$. Thus $n - 1$ is even and has at least three factors. 1, 2, and itself. So $\sigma(n - 1) = 1 + 2 + n - 1$.

$\sigma(n) = 1 + p, \sigma(n - 1) = n + 2 = p + 2$. $2 + p$ is greater than $1 + p$.

Problem 17:

$$x^2 - y^2 = N \leftrightarrow (x + y)(x - y)$$

Any pair factors will satisfy, so the number of solutions as of now is $d(n)$. Of course, this is x^2 , so the negative of a solution will also work. This is double the solutions or $2d(n)$.

Problem 20:

Let d be the divisors of n . $d_1, d_2 \dots d_x$. Every d has a corresponding d that when multiplied equals n . It then follows that the number of divisors, including 1 and n , must be even or else a $dx = n$.

But n is already a factor. It then follows that the amount of n 's that are multiplied is $\frac{d(n)}{n}$ as each pair equals one $n * n \dots * n$ for a $\frac{d(n)}{2}$ of times equals $n^{d(n)/2}$.

Chapter 8

Problem 7:

$$\sigma(\sigma(n)) = 2^n$$

If $n = 2^k$, then $\sigma(n) = 2^{k+1} - 1$. Since $2^{k+1} - 1$ is prime

$$\sigma(2^{k+1} - 1) = 2^{k+1} - 1 + 1 = 2^{k+1} = 2 * 2^k = 2 * n * \sigma(2^{k+1} - 1) = \sigma(\sigma(n)).$$

Problem 10:

Yes, first realize that this problem is simply asking if an infinite amount of triangular numbers can be even perfect numbers. From the previous problem:

$$2^p(2^p - 1)/2 = \text{an even triangular number.}$$

So any number in that form, as long as $2^p - 1$ is prime, will be an even perfect number and a triangular number.

Problem 11:

Since all the numbers are prime,

$$\sigma(a) = (2^{e+1} - 1)(3 * 2^e)(3 * 2^{e-1}) \text{ and } \sigma(b) = (2^{e+1} - 1)(3^2 * 2^{2e-1})$$

$\sigma(a) - a = b$, and $\sigma(b) - b = a$, so the numbers are amicable.

Problem 12:

$$\sigma(2p(2p - 1)) = \sigma(2p) * \sigma(2p + 1) = 3(p + 1) * 2(p + 1)$$

$$n = 2p^2 + 10p + 6$$

This is less than $4p^2 = 2p$ for all $p > 3$.

Problem 13:

A perfect number $= 2^{p-1}(2^p - 1)$ where $2^p - 1$ is a prime. So $2^p - 1 \equiv 3, 7 \text{ modulo } 10$.

Being congruent to 2 implies divisibility by 2. For reason not covered here, $n \equiv 1$ implies that p is a multiple of 5. $2^p - 1 \equiv 3$ implies that $2^{p-1} \equiv 2 \text{ modulo } 10$. $3 * 2 = 6$ and $7 * 4 = 28 \equiv 8 \text{ modulo } 10$. This covers the only two possible cases.

Chapter 9

Problem 7:

If n is odd then

$$\sigma(4n) = \sigma(2^2) * \sigma(n) = 2\sigma(n)$$

Problem 8:

None do: If one did,

$$n = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k}, \text{ so}$$

$$\sigma(n) = p_1^{e_1-1}(p_1 - 1) \dots p_k^{e_k-1}(p_k - 1) =$$

$$n(1 - 1/p_1)(1 - 1/p_2) \dots (1 - 1/p_k) < n < 2n$$

Problem 10:

$$\Sigma\sigma(p) - \Sigma\phi(p) = \Sigma d(p)$$

$\sigma(p)$ is all numbers below p . Removing all numbers relatively prime to p , you must be left over with only factors of p .

Problem 11:

$$ar + ms = 1$$

If $(a, m) = 1$ and $a \equiv b \pmod{m}$ then

$$(b, m) = 1$$

Since a is congruent to b modulo m

$$a = mk + b, \text{ so:}$$

$$ar + ms = 1 \rightarrow mkr + br + ms = 1 \rightarrow br + m(kr + s) \rightarrow br + ms' = 1$$

Which implies that $(b, m) = 1$

Problem 12:

$$x \equiv ca^{\sigma(m)-1} \pmod{m}$$

Multiplying the entire equation by a ,

$$ax \equiv ca^{\sigma(m)} \equiv c \pmod{m}$$

Problem 14:

$$16 = 4 * 4 = 8 * 2 = 16 * 1$$

By finding solutions based on those relationships, one can easily find 4 solutions. Examples are 17, 32, 34, 40.

Problem 15:

$$4 = 2^2 = 4 * 1$$

Solutions are 8, 5, 12, 10.

Assume that some x has the property of $\phi(x) = 4$.

$$x = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k}$$

Every prime greater than 5 has $\phi(p) > 5$, so the greatest prime x can possess is 5. So x must be of the form $2^x * 3^y * 5^z$. The largest x is 3, and the largest possible power of 3 and 5 is 1.

Therefore, there are only $3 * 2 * 2 = 12$ amount of numbers possible, and it is not too tedious to check through all of them to find that the before mentioned numbers are the only ones who hold the property.

Problem 17:

$$m = 2a$$

$$n = 2b$$

From problem 16 we know that the difference can be shown as $p_1(p_1 - 1) > (p_1 - 1)^2$.

Since $d = 2, p_1 = 2$,

$2(1) > 1^2$ by a factor of multiplication of 2. So $\sigma(mn) = 2\sigma(m)\sigma(n)$ if $(m, n) = 2$.

Problem 19:

Since $n - 1, n + 1$ are primes it follows that $n = 6x$. Say that $d(n)$ shows the number of divisors of n . At the least, $d(n) = n/2 + n/3$ or $d(n) \geq 2n/3$. Since $2n/3 + n/3 = n$ it follows that the number of numbers who do not divide n is $\sigma(n) \leq n/3$.

Problem 20:

$$14 = 7 * 2$$

No prime has the property of $\phi(p) = 7$.

$$\phi(n) = (p_1^{e_1-1})(p_1 - 1) \dots (p_k^{e_k-1})(p_k - 1) = 14$$

$$\text{So one of } (p_x^{e_x-1})(p_x - 1) = 7$$

But $7 = 7*1$, so one of $p_x^{e_x-1}, (p_x - 1)$ must equal one. If $p_x - 1 = 1$ then $p_x = 2$. But then

$2^k = 7$, which is not possible. The only other case is $p_x = 8$. But then, the formula yields 28 not 7, leaving no possibility to equal 14.

Chapter 10:

Problem 9:

Since $1 \equiv a^t \pmod{m}$

$$a^{t-1} + a^{t-2} \dots + 1 \equiv a^{t-1} + a^{t-2} \dots + a^t$$

$$\text{Say that } A = a^t + a^{t-1} \dots + a$$

$$A * a = a^{t+1} + a^t \dots + a^2$$

$$Aa - A = A(a - 1) = a^{t+1} - a = a(a^t - 1) \leftrightarrow A = \frac{a(a^t - 1)}{a - 1}$$

We know that $a^t - 1 \equiv 0 \pmod{p}$ and that $a - 1$ divides $a^t - 1$ because $x - y$ divides $x^n - y^n$.

Thus the sequence A is always divisible by p .

Problem 10:

From Lemma 1, $(k, p - 1) = 1$. This is only possible if k is odd, as $p - 1$ is even.

Problem 11:

$g \equiv h^k$ for some odd k . So $gh \equiv h^{k+1} \pmod{p}$. But h^{k+1} is only a primitive root if $(k + 1, p - 1) = 1$

However, these are both even, so this must be false.

Problem 12:

Yes, $ghk \equiv h^{k+1}h^y \equiv h^{k+y+1}$ where $k + y + 1$ is odd. From Lemma 1, this is a primitive root of p .

Problem 13:

If $(a + 1)^6$ is congruent to 1 then $(a + 1)^3$ is congruent to -1.

$$(a + 1)^3 = a^3 + 3a^2 + 3a + 1$$

$$3(a^2 + a) + 2$$

From Problem 9, we know that $a^2 + a$ is congruent to -1 modulo p . So:

$$3(a^2 + a) + 2 \equiv -3 + 2 \equiv -1.$$

Problem 14:

We know that $a^p \equiv -1 \pmod{p}$, and the order of a divides p , so the order is either 1 or p . If it is one then q divides $a - 1$. If the order is p , then p divides $\phi(q)$ so $px + 1 = q$ and we know $x = 2k$ because $q - 1$ is even.

Problem 15:

$$a^2 \equiv -1 \pmod{p}$$

$$(a + 1)^2 = a^2 + 2a + 1 \equiv 2a$$

$$(2a)^2 = 4a^2 \equiv -4 \pmod{p}$$

Problem 18:

$$g^{k+1} \equiv g^k + 1 \pmod{p} \rightarrow g^k(g - 1) - 1 \equiv 0 \pmod{p}$$

And from Lemma 2 we know that this has at most k solutions.

Problem 19:

$$g^{k+2} \equiv g^{k+1} + 1 \equiv g^{k+2} \pmod{p} \rightarrow g^{k+1}(g - 1) - 1 \equiv g^k(g - 1) - 1. \text{ So:}$$

$$g^{k+1} \equiv g^k \text{ but then } g^{k+1} + 1 \text{ is not congruent to } g^{k+2}.$$

Problem 20:

$$\text{If } p \text{ is a prime then } (p - 1)! \equiv -1 \pmod{p}$$

$$(p - 1)! - py = 1 \text{ which we know to be true.}$$

$$g^{\phi(m)} * g^{\phi(m)+1} \dots * g = g^{\frac{\phi(m)*\phi(m)+1}{2}}$$

$$(g^{\frac{\phi(m)}{2}})^{\phi(m)+1} \equiv -1^{\phi(m)+1} \equiv -1^{2k+1} \equiv -1.$$

Chapter 11:

Problem 13:

If p is of the form $8k + 7$ or $+1$, then $(p^2 - 1)/8 = 2k_1$ which yields 1. If it is of the form $8k + 3$ or $+5$, then $\frac{p^2-1}{8}$ is odd and yields -1.

Problem 14:

First realize that one of p, q is of the form $4k + 1$. Then $(p/q)(q/p) = 1$. If this is not true, then $(p/q)(q/p) = -1$. If one of $p, q = 4k + 1$, then:

$$(-1)^{\frac{(4k+1-1)(q-1)}{4}} \rightarrow (-1)^{k(q-1)}$$

$$\text{And since } q - 1 \text{ is even, } (-1)^{k(q-1)} \rightarrow 1$$

$$\text{If not, } (-1)^{\frac{(4k_1+2)(4k_2+2)}{4}}. \text{ Which is an odd number, so:}$$

$$(-1)^{\frac{(4k_1+2)(4k_2+2)}{4}} \rightarrow -1.$$

Problem 15:

Because 7 is congruent to 3 modulo 4, it does not have a solution.

$$a^2 + 1 \equiv 0 \pmod{7}$$

$$a^2 \equiv -1 \pmod{7}$$

$$(-1/7)$$

Problem 16:

$ab \equiv 1 \pmod{p}$. Raising this to the $(p-1)/2$ power:

$a^{(p-1)/2} * b^{(p-1)/2} \equiv 1$. Since a is a quadratic residue, $a^{(p-1)/2} \equiv 1$. So:

$1 * b^{(p-1)/2} \equiv 1 \pmod{p}$ and so $b^{(p-1)/2} \equiv 1 \pmod{p}$.

This is the definition of a quadratic residue.

Problem 18:

So we are evaluating

$$(4k+1)(8k+3) \rightarrow$$

$$(8k+3/4k+1), \text{ and } 8k+3-4k-1-4k-1=1$$

So $(1/4k+1)$, this is of course true because $1^{4k+1-1=4k} = 1$

Problem 19:

$ab \equiv r \pmod{p}$. Raised to the power $(p-1)/2$:

$1 * b^{(p-1)/2} \equiv r^{(p-1)/2}$. Since r is a quadratic residue this goes to

$b^{(p-1)/2} \equiv 1$. This is again the definition of a quadratic residue.

Problem 20:

$$(p/q) = (q/p) = (-4a/p) = (-1/p)(2^2/p)(a/p) = 1 * 1 * (a/p) = (a/p)$$

Chapter 12:

Problem 4:

$8k+7$ when divided by 2 leaves a remainder is 1. The greatest integer less than this is some $2k$

This of course makes $(-1)^g = (-1)^{2k} = 1$, so $2^{(p-1)/2} \equiv 1 \pmod{p}$.

$(p-1)/2 = 83$. Multiplying by 2,

$$p-1 = 166 \rightarrow p = 167.$$

167 is of the form $8k+7$

Problem 5:

If $p = 4k + 3$

$$(p/q) \rightarrow - (q/p) = - (1/p) = (-1/p)$$

If $p = 4k + 1$

$$(p/q) = (q/p) = (1/p)^{-1} = (-1/p)$$

Problem 6a:

$$n^2 \equiv -1$$

$$(-1/p)$$

So prime of the form $4k + 1$.

Problem 6b:

If n, p have some common factor other than 1, it is trivially true. If not:

$$n^2 \equiv -1 \rightarrow n \equiv -1 \rightarrow n^2 \equiv 1 \pmod{p}$$

This is true for all primes.

Problem 6c:

$$n^2 + 2n + 2 \rightarrow (n + 1)^2 \equiv -1 \pmod{p}$$

All primes of the form $4k + 1$ will divide.

Problem 7a:

$$n^2 \equiv a \pmod{p}$$

$$p = 4k + 3$$

$$(a/p) = 1$$

$$(p - a/p) \rightarrow (-a/p) \rightarrow (-1/p)(a/p)$$

$$(a/p) = 1 \text{ and } (-1/p) = -1, \text{ so}$$

$$(-1/p)(a/p) = -1$$

Problem 8:

A sum of squares equals $\frac{(n-1)(n)(2n+1)}{6}$. So the sum of least residues squared up to p sum to

$\frac{(p-1)(n)(2n+1)}{6}$ which p divides. If a number divides a square, then it also divides the root.

Chapter 13:

Problem 16:

Say that there is a number k that does not hold this property. After subtracting all powers of three times some e , the difference is either one or two. $1 = 1 * 3^0$, and $2 = 2 * 3^0$. These are the only two cases possible. If anything above 2 is left, then a $3^x d_x$ can be subtracted to bring it back down to a least residue of three.

Problem 18:

$$b * 2^{3s} \text{ but } b = (2 + 2 + 2 \dots + 1). (2 + 2 + 2 \dots + 1) = 2^{3s+1} + 2^{3s} \dots + 2.$$

$$2^y + 2^{y-1} \dots + 2^{3y-x} = b * 8^s.$$

Now all that is left is to show that this is the base 2 showing of b . Dividing this sequence by 2^3 this sequence is the base 2 showing of b .

Problem 19:

The base case is true, and assume all numbers up to n are true. Take an $n + 1$.

$n' = n + 1 - e! * d_e$ where $e! * d_e$ is the largest factorial that fits into $n + 1$.

$n' < n$, so n' has a unique factorization, adding back the $e! * d_e$, this gives the unique factorization of $n + 1$.

Problem 20:

$$4b + 5 = x^2$$

$$5b + 5 = (x + 1)^2$$

$$x^2 - 4b = 5 \text{ so}$$

$$5b + x^2 - 4b$$

$$b + x^2 = x^2 + 2x + 1 \rightarrow b = 2x + 1$$

$$4(2x + 1) + 5 = x^2$$

$$8x + 4 + 5 = x^2 \rightarrow x^2 - 8x - 9 = 0 \rightarrow (x - 9)(x + 1) = 0$$

$$x = -1 \text{ or } 9$$

So b equals -1 or 19. Plugging this back in, base 81 and 100 will do.

Chapter 14

Problem 7a:

6^x is always a sum of powers of $d * 12^n$. Since $6^2 = 3 * 12^1$, all 6^x are multiples of 12. So there will no d that usually creates the last digit. The last digit is then 0.

Problem 7b:

10^x always has $10^x \equiv 4 \pmod{12}$

$$10^2 \equiv 4$$

$$10 * 10^2 \equiv 4$$

$$10^2 * 10^2 \equiv 4$$

And every power of 10 can be made from these operands.

Problem 7c:

Using the same logic as that in 7b, one can construct a table for all numbers.

Problem 8:

$$x \equiv 3 \pmod{12}$$

$$x = 3(4k + 1)$$

And you can see how this works for all factors of 3. 0 is trivial.

Problem 9:

Using the same logic as Problem 8,

$$x \equiv 4k \pmod{12}$$

$$x = 4(3k_2 + k_1)$$

$$x \equiv 6k \pmod{12}$$

$$x = 6(2k_2 + k_1)$$

Problem χ :

$$12 \equiv 1 \pmod{11}$$

So if a number $a_1, a_2 \dots a_k$ has digits that add to a multiple of 12, then

$12^k a_k + 12^{k-1} a_{k-1} \dots + a_1 \equiv a_k + a_{k-1} \dots + a_1$. Which we have established sums to a multiple of 11.

Chapter 15:

Problem 5:

$$1/n = d_1/b + d_2/b^2 + d_3/b^3 \dots + d_k/b^k$$

Every prime divisor of n is a prime divisor of b .

Multiply the entire sequenced by b^k . Then:

$$b^k/n = b^k(d_1/b + d_2/b^2 \dots + d_k/b^k)$$

This is strictly an integer, so n divides b .

Problem 6:

$$b^k/n = b^k(d_1/b + d_2/b^2 \dots + d_k/b^k)$$

So the sequence equals a set integer. This can only happen if the decimal expansion terminates. If it doesn't, and the expansion equals an integer at some point, then there is always another term to make it a fraction.

Problem 14:

If x is not a square, then $\sqrt{x}$ is irrational.

Proof:

$$\sqrt{x} = \frac{a}{b} \rightarrow x = \frac{a^2}{b^2}$$

This implies the x itself is a square.

$$\frac{1}{7^x} = \frac{1}{7^{x(x+1)}} = i_x$$

$$i_1 + i_2 \dots + i_k = \frac{a}{b} = \frac{7^{i_x}}{7^{i_y}} \text{ where } i_y > i_x.$$

$$\frac{7^{i_x}}{7^{i_y}} \rightarrow \frac{1}{7^{i_y - i_x}} = \frac{1}{7^x}$$

This is irrational.

Problem 15:

The base case is true, and assume for up to n . Say that $n + 1 = .a_1 a_2 \dots$

Subtract the largest $.a_1$ from $n + 1$. This new number is less than n , so has a terminating expansion. Adding back the $.a_1$, you get the decimal expansion of $n + 1$.

Chapter 16:

Problem 5:

If there was a greater divisor between any two terms, it must also divide the third, so $(a, b) \neq d$. Furthermore, d must divide c .

Problem 6:

$ab = c$ but none of a or b have common factors so each $p_{e_k}^{x^y}$ is distinct. Therefore, c^n will have the same power as a and b .

Problem 7:

Since $(a, b) = d$, there is at least one prime factor that they share. These primes will not have the same power of a, b .

Problem 8:

$$a = 2mn$$

$$b = m^2 - n^2$$

$$2mn(m^2 - n^2) = 2h \rightarrow 2m^3n - 2mn^3 = 2h.$$

$$m^3n - mn^3 = h \text{ which is not of the form } m^2 + n^2 = h \text{ because } (m, n) = 1.$$

Problem 9:

$$a^2 + b^2 = c^2$$

$a^2; b^2$ are congruent to 0, 1, 4 (mod 5).

0 + 0; 1 + 1; 4 + 4 are all not possible because c^2 is a square and thus congruent to 0, 1, 4 modulo 5. The only other combinations lead to a multiple of 5 being present.

Problem 10:

We assume a is even, so we already have to 4 part of the $4 * 3$. One of b^2, c^2 must contain 3.

$$c^2, a^2, b^2 \equiv 0, 1 \pmod{3}$$

If $a \equiv 0$ then $a^2 + b^2 \equiv 1$. If $a \equiv 1$, then $1 + x \equiv y$. But $y \equiv 0, 1$. To be congruent to 0 x^2 must be congruent to 2 but that is impossible. So $x^2 \equiv 0$. All of these cases lead to a factor of three, which paired with the *4 leads to a factor of 12.

Problem 11:

The legs are a multiple of 12 while the hypotenuse is a multiple of 5. Therefore all sides multiplied together are a multiple of 60.

Problem 13:

$$ab/2 = 2(a + b + c)$$

$$ab/4 - a - b = c$$

$$(ab/4 - a - b)(ab/4 - a - b) \leftrightarrow a^2b^2/16 - a^2b/4 - ab^2/4 - a^2b/4 + a^2 + ab - ab^2 + ab + b^2 \rightarrow a^2b^2/16 - a^2b/2 - ab^2/2 + 2ab = 0. [16]$$

$$a^2b^2 - 8a^2b - 8ab^2 + 32ab = 0 \rightarrow ab - 8a - 8b + 32 = 0$$

$$(a - 8)(b - 8) = 32$$

Must be $32*1$ because all others involve 2 evens. Thus $a = 40, b = 9, c = 41$.

Problem 14:

$$3(a + b + c) = ab/2$$

$$c = ab/6 - b - a. \text{ Squaring}$$

$$a^2b^2/36 - a^2b/3 - ab^2/3 + a^2 + b^2 + 2ab. \text{ So:}$$

$$a^2b^2/36 - a^2b/3 - ab^2/3 = 2ab = 0. \text{ Multiplying by } [36/ab]$$

$$ab - 12a - 12b + 72$$

$(a - 12)(b - 12)$. Trying different factoring of 72, the only possible triangles are:

(36, 15, 39) and (84, 13, 85)

Problem 15:

$$(x)(x + 1)(x + 2) = 60k$$

Since one is divisible by 5, and one by three, this is only possible if $x = 3$, thus (3, 4, 5) is the only triangle.

Problem 16:

$$a^2 + b^2 = c^2 \text{ where } b = a + x, c = a + 2x$$

$$abc = 60k$$

$$a^2 + a^2 + 2ax + x^2 - a^2 - 4ax - 4x^2$$

$$a^2 - 2ax - 3x^2 = 0$$

$$(a - 3x)(a + x) = 0 \rightarrow a = 3x$$

$$\text{Thus } b = 4x, c = 5x.$$

Problem 17a:

$$(2n + 1, 2n(n + 1), 2n(n + 1) + 1$$

Problem 17b:

$$a = (2n + 1)^2$$

$$a^2 = 4n^2 + 4n + 1 = 2b + 1$$

$$2n^2 + 2n = b = 2n(n + 1)$$

$$a^2 + b^2 = (b + 1)^2 \rightarrow a = 2b + 1.$$

Problem 19:

$$n^2 + (n + 1)^2 = 2m^2$$

So 2 divides $n, n + 1$. This is not possible.

Chapter 17:

Problem 2:

Because one is unable to define integer numbers above and below n in order to create a lesser representation.

Problem 3:

$$\text{No, } x^8 + y^8 = z^8 \rightarrow x^{2^4} + y^{2^4} = z^{4^2}$$

Problem 4:

Suppose that there is a smallest solution.

3 divides x , so $x^3 = 27x^3$.

Then $27x^3 + 3y^3 = 9z^3$.

Dividing by three yields a smaller answer, which is a contradiction.

Problem 5:

No, you can create a smaller answer by dividing by p^x .

Problem 6:

No, use the same logic as 5.

Problem 7:

No, the x^n will always be divisible by a higher power of p , and you can always create a smaller solution.

Problem 8:

$$x = 2a$$

$$y^2, z^2 \equiv 1 \pmod{4}$$

$$y^2 + z^2 = 2xyz - x^2 \rightarrow y^2 + z^2 = 4ayz - 4a^2 \rightarrow 4(ayz - a^2)$$

But $y^2 + z^2 \equiv 2 \pmod{4}$. This is a contradiction.

Problem 9:

Any $b = 2b$ will yield no solutions as a contradiction of modulus will come up.

Problem 10:

x^2, y^2 must both be odd.

$x^2 + y^2$ is even, but $x^2 y^2$ is odd. Since they are supposedly equal, this cannot be true.

$$(xy)^2 \equiv 0, 1 \pmod{3}$$

The only possible way to make this happen is if $x, y, z \equiv 1 \pmod{3}$.

$$x^2 = 3a + 1$$

$$y^2 = 3b + 1$$

$$z^2 = 3c + 1$$

$$3a + 3b + 3c + 3 = (3b + 1)(3a + 1)$$

Obviously the left side is divisible by 3, while the right hand side is not. This is a contradiction and gives our answer.

Chapter 18:

Problem 4:

The first representation I found is $63^2 + 22^2$. The second is $58^2 + 33^2$. Changing the a, b, c, d order yields different representations.

Problem 5:

$$x^2; y^2 \equiv 0, 1 \pmod{4}$$

With these two numbers it is impossible to make a number that is congruent to 3 modulo 4.

Problem 6:

They can be congruent to 1,2,3,4,5,6,7,8 modulo 9. You can easily find this by squaring every number and trying different combinations.

Problem 7:

The common factor $(m, n) = d$ is a square.

$$s^2 = d^2 a^2$$

$$r^2 = d^2 b^2$$

$$s^2 = ma^2 \text{ and } r^2 = mb^2 \text{ so } n/m \rightarrow$$

$$(s^2 + r^2)/m = (ma^2 + mb^2)/m \rightarrow a^2 + b^2$$

So yes it is true.

Problem 8:

If $mn \equiv 3 \pmod{4}$ then it is not representable. If it is congruent to 1 then it is representable.

Both $m, n = 4k_x + 1$. So:

$$(m)(n) = (4a + 1)(4b + 1) = 16ab + 4a^2 + 4b^2 + 1 \equiv 1 \pmod{4}.$$

So it is representable as the sum of two squares.

Problem 9:

n being even assures that $2n + 1 \equiv 1 \pmod{4}$.

Dividing by the largest square leaves only odd numbered powers. If the residue is divisible by a $4k - 1$ then that divisible prime is of the form $4k + 3$ and is not representable. This is equivalent to the theorem.

Problem 10:

Every integer of the form $4k + 1$ can be created from these instructions, and none of the integers of the form $4k + 3$ can. This is then equivalent to the theorem.

Chapter 19:

Problem 5:

x^2, y^2 must be congruent to one modulo 4. Any integer of that form will do. An example is $2 + 23^2 - 19^2 = 17 * 10$

Problem 7:

The squares are congruent to 1, 0, 4 (mod 8). However, congruency to 4 is not possible for a square, as a square containing 4 also contains 16. The only way the squares are even is if the squares are congruent to:

$1 + 1 + 1 + 0$ or $1 + 0 + 0 + 0$ modulo 8.

In the first case $x^2 + y^2 + z^2 + w^2 = 8a \equiv 3$. This is untrue. Similarly for the second case, 1 is not congruent to 0. Since these are the only two cases in which all of the squares are not even, they must all be even.

Problem 10:

Case 1:

x, y, z are all odd.

$t^2 \equiv 0$ while $x^2 + y^2 + z^2 \equiv 1 \pmod{2}$.

These do not match up, so they cannot all be odd.

Case 2:

x is even, y^2, z^2 are odd. Then

$z^2 + y^2 \equiv 0 \pmod{4}$. Two squares cannot be congruent to 4, as then at least one must be congruent to 2 or 3.

Chapter 20:

Problem 7b:

$$x^2 + 2axy + by^2 = 1$$

$a^2 < b$ implies that $x^2 + 2axy + by^2$ will always be greater than one because all of the terms are positive.

Problem 7c:

Same logic as problem 7b.

Problem 8a:

$$(m - n)^2 - 2n^2 = 1$$

$$m^2 - 2mn + n^2 - 2n^2 = 1$$

$$c - a - 2n^2 = 1$$

$$c = 2n^2 + a + 1 \rightarrow 2n^2 + m^2 - n^2$$

$$c = m^2 - n^2. \text{ Which is true.}$$

Working backwards

$$c = 2n^2 + m^2 - n^2 \rightarrow 2n^2 + a + 1 \rightarrow c - a - 2n^2 = 1 \rightarrow$$

$$m^2 - 2m + n^2 - 2n^2 = 1 \rightarrow (m - n)^2 - 2n^2 = 1$$

Problem 9a:

$$x^2 = \sqrt{3a(a-1)(a)(a+1)} \rightarrow \sqrt{3a^2(a-1)(a+1)} \rightarrow \sqrt{3a^2(a^2-1)} =$$

$$x^2 = a\sqrt{3(a^2-1)}. \text{ So } 3(a^2-1) \text{ must be a square.}$$

Problem 10:

$$3((2a+1)^2 - 4) \neq x^2$$

$$(2a+1)^2 - 4 = 9x^2 \rightarrow 4a^2 + 4a - 3 = 9x^2 \rightarrow 4(a^2 + a) = 9x^2/4 + 3/4$$

So x^2 must be congruent to 1 modulo 4. But then a^2 must be congruent to 0, which is a contradiction.

Problem 15:

Doing some basic congruences, one can find out that the starting square must be a multiple of 27, and that the starting square on the right side must be odd. It then only takes a few guesses to find that:

$$108^2 + 109^2 + 110^2 = 133^2 + 134^2$$

Chapter 21:

Problem 1a:

2^{1979} is the largest such power. This can easily be found using the methods found in the text.

Problem 1b:

This is impossible. Consider:

$$n/2 + n/4 \dots + n/2^x = n.$$

Multiply this sequence by the 2^{x-1} . Then

$2^{x-2} + 2^{x-3} \dots + 1/2 = 2^{x-1}n$. This is a sequence of integers plus a fraction equaling an integer, which is not possible.

Problem 3:

$n!$ does not contain p , and $(2n)!$ can only contain one p because $2p > 2n$.

Say that p^2 is in the prime factorization of $2n$. Then $p + a = 2n$ so $p^2 = p^2 + 2a + 2ap + a^2 = np + a^2$ and $p > 2$ so $np + a^2 > 2n$ so p^2 cannot be in the prime factorization of $2n$.

Problem 4:

p appears in the bottom fraction twice.

$2p < 2n$ but $3p > 2n$ so 2 multiples of p appear at the top. These cancel out, and leave no p in the prime factorization.

Problem 5:

$$\pi(x)\ln(x) \leq ax$$

$$\pi(x) > x, \text{ so:}$$

$$a > \ln(x)$$

$$\ln(x) > \ln(n), \text{ so}$$

$$a > \ln(x) > \ln(n) \text{ Thus in}$$

$$p_n a > n \ln(n), p_n > n \text{ and } a > \ln(n).$$

With these two inequalities, it is easy to see that the equality is true.